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SEMICONDUCTOR PACKAGE

Abstract

A semiconductor package for detecting a pressure of a pressure medium flowing through a pipe. The semiconductor package includes: a case body having a pressure detection chamber; a pressure sensor chip housed in the pressure detection chamber and configured to convert a pressure applied thereto into an electrical signal; a lead frame insert-molded in the case body and electrically connected to the pressure sensor chip; and a pressure intake part having a first end protruding externally from a bottom of the case body, and a second end spatially connected to the pipe. The case body has a channel that extends from the first end of the pressure intake part and passes through a vicinity of the lead frame, to thereby guide the pressure medium from the first end of the pressure intake part through the channel to the pressure detection chamber, to apply the pressure to the pressure sensor chip.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2024-028219, filed on Feb. 28, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] 1. Field of the Invention Embodiments of the disclosure relate to a semiconductor package.

[0003] 2. Description of the Related Art

[0004] International Publication No. WO 2017/212800 describes a technology that is a pressure sensor used to measure intake pressure or exhaust pressure of an internal combustion engine and has, in an outer case: a pressure measuring chamber into which a gas to be measured is fed, a sensor chip facing the pressure measuring chamber, a sensor supporting member having a supporting surface that supports the sensor chip, a temperature maintaining chamber facing a back surface (surface opposite to the supporting surface) of the sensor supporting member, and a gas channel coupling the temperature maintaining chamber and the pressure measuring chamber, the gas to be measured is fed into the temperature maintaining chamber from the pressure measuring chamber via the gas channel and thereby maintains the temperature of the sensor chip from both sides thereof, whereby condensation is suppressed around the sensor chip.

SUMMARY OF THE INVENTION

[0005] According to an embodiment of the present disclosure, a semiconductor package for detecting a pressure of a pressure medium flowing through a pipe, the semiconductor package includes: a case body having a pressure detection chamber; a pressure sensor chip housed in the pressure detection chamber and configured to convert a pressure applied thereto into an electrical signal; a lead frame insert-molded in the case body and electrically connected to the pressure sensor chip; and a pressure intake part having: a first end protruding externally from a bottom of the case body, and a second end spatially connected to the pipe. The case body has a channel formed in the bottom thereof coupling the pressure detection chamber and the pressure intake part, the channel extending from the first end of the pressure intake part and passing through a vicinity of the lead frame, to thereby guide the pressure medium from the first end of the pressure intake part to the pressure detection chamber, such that the pressure medium applies the pressure to the pressure sensor chip in the pressure detection chamber.

[0006] Objects, features, and advantages of the present invention are specifically set forth in or will become apparent from the following detailed description of the invention when read in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a cross-sectional view schematically depicting a structure of a semiconductor package according to a first embodiment.

[0008] FIG. **2** is a plan view schematically depicting a layout when an inside of a case body depicted in FIG. **1** is viewed from a cover body.

[0009] FIG. **3** is a cross-sectional view schematically depicting a structure of a semiconductor package according to a second embodiment.
[0010] FIG. **4** is a cross-sectional view schematically depicting a structure of a semiconductor package of a reference example.
[0011] FIG. **5** is a plan view schematically depicting a layout when an inside of a case body depicted in FIG. **4** is viewed from a cover body.

DETAILED DESCRIPTION OF THE INVENTION

[0012] First, problems associated with the conventional technology are discussed. In International Publication No. WO 2017/212800, only a space bordered by a cover of the outer case can be heated and a temperature of a connector pin functioning as a heat sink for externally dissipating heat around the sensor chip is lower than a temperature of components in the outer case. Thus, in a vicinity of a connector between the connector pin and a connector terminal of the sensor supporting member, vapor contained in the gas to be measured is naturally cooled, generating condensate in the pressure measuring chamber. This condensate may corrode wiring and/or the connector terminal of the sensor supporting member, causing disconnection and/or short-circuit.
[0013] An outline of an embodiment of the present disclosure is described. (1) A semiconductor package according to an embodiment of the disclosure is a semiconductor package for detecting pressure of a gas flowing through a pipe and is as follows. A case body has a pressure detection chamber to which a pressure medium is sent. A pressure sensor chip is housed in the pressure detection chamber and is configured to convert pressure applied by the pressure medium into an electrical signal. A lead frame is insert molded to the case body and is electrically connected to the pressure sensor chip. A pressure intake part protrudes externally from a bottom of the case body and is spatially connected to the pipe. The gas, which constitutes the pressure medium, flows into the pressure intake part from the pipe. The pressure intake part has a closed end closed by the bottom of the case body and an open end spatially connected to the pipe. Inside the bottom of the case body, a channel coupling the pressure detection chamber, and the pressure intake part is formed passing through a vicinity of the lead frame from the closed end of the pressure intake part to the pressure detection chamber.
[0014] According to the disclosure above, the lead frame may be actively heated. As a result, the temperature of the lead frame may be made higher than the temperature of the pressure sensor chip and thus, even when vapor is contained in the pressure medium, condensation may be suppressed in a vicinity of metal members forming an electrical connector between the lead frame and the pressure sensor chip.
[0015] (2) Further, in the semiconductor package according to the disclosure, in (1) above, the case body has a space different from the pressure detection chamber and the channel may pass through the space, from the closed end of the pressure intake part to the pressure detection chamber.
[0016] According to the disclosure above, the space of the case body, different from the pressure detection chamber may be used as a space for naturally cooling the pressure medium or may be used for heat transfer to the lead frame.
[0017] (3) Further, in the semiconductor package according to the disclosure, in (2) above, the channel may include a first channel that passes through the vicinity of the lead frame and couples the pressure intake part and the space, and a second channel that couples the space and the pressure detection chamber.
[0018] According to the disclosure above, heat from the pressure medium that flows into the space of the case body, different from the pressure detection chamber is dissipated to the outside via an outer wall (side surface and top) of the case body thereby facilitating cooling of the pressure medium naturally. As a result, the amount of vapor contained in the pressure medium decreases enroute to the pressure detection chamber and thus, condensation in the pressure detection chamber may be suppressed.
[0019] (4) Further, in the semiconductor package according to the disclosure, in (2) or (3) above,

the space may be bordered by the top of the case body. According to the disclosure above, in the space of the case body,

[0020] different from the pressure detection chamber, heat may be dissipated from the pressure medium to the outside via the top of the case body.

[0021] (5) Further, in the semiconductor package according to the disclosure, in (3) above, the second channel may be formed near the side surface of the case body.

[0022] According to the disclosure above, in the second channel, heat may be dissipated from the pressure medium to the outside via the side surface of the case body.

[0023] (6) Further, in the semiconductor package according to the disclosure, in (1) above, the space is formed inside the bottom of the case body, in the vicinity of the lead frame. The channel may include the first channel that couples the pressure intake part and the space, and the second channel that couples the space and the pressure detection chamber.

[0024] According to the disclosure above, the lead frame may be actively heated by the pressure medium that flows into the space of the case body, different from the pressure detection chamber.

[0025] (7) Further, in the semiconductor package according to the disclosure, in (6) above, the space may face the lead frame in a direction orthogonal to the bottom of the case body.

[0026] According to the disclosure above, heating of the lead frame by the pressure medium that flows into the space of the case body, different from the pressure detection chamber, is facilitated.

[0027] Findings underlying the present disclosure are discussed. A structure of an on-board automotive sensor package for measuring the pressure of exhaust of an internal combustion engine (engine) is described as a semiconductor package of a reference example. FIG. **4** is a cross-sectional view schematically depicting the structure of the semiconductor package of the reference example. FIG. **5** is a plan view schematically depicting a layout when an inside of a case body depicted in FIG. **4** is viewed from a cover body. A semiconductor package **110** of the reference example depicted in FIGS. **4** and **5** has a pressure sensor **101** and a pressure intake part **102** and is an on-board automotive sensor package installed to exhaust system piping (exhaust pipe, not depicted) of the engine. The pressure sensor **101** is equipped with a pressure sensor chip **121** mounted in a pressure detection chamber **112** inside a case body **111**.

[0028] The pressure intake part **102** is a resin pipe (piping made of resin) integrally molded with the case body **111** and connects the pressure detection chamber **112** and the exhaust pipe. The pressure intake part **102** is a channel for a pressure medium **104** that is sent to the pressure detection chamber **112**. A first open end **102***a* of the pressure intake part **102** is connected to the pressure detection chamber **112** at a bottom (side facing the exhaust pipe) **111***a* of the case body **111**. A second open end **102***b* of the pressure intake part **102** is inserted into an opening formed in the exhaust pipe of the engine, whereby the semiconductor package **110** is directly attached to the exhaust pipe. The pressure intake part **102** connects the pressure detection chamber **112** and the exhaust pipe spatially in a straight line over a shortest distance therebetween.

[0029] The case body **111** is a resin molded product in which a lead frame **113** is insert molded. The case body **111** has, in a cross-sectional view, a substantially concave shape opened in one portion and the opened portion is covered by a cover body **115**. A space (space bordered by an inner wall (the bottom **111***a* and sidewalls) of the case body **111** and a cover body **115** (top)) inside the case body **111** is spatially isolated from the pressure detection chamber **112** and a wiring chamber **114** by a housing container body **122**. The pressure detection chamber **112** is spatially connected to the exhaust pipe linearly by the pressure intake part **102**. The wiring chamber **114** is an enclosed space shielded from the outside air by the case body **111** and the cover body **115** and shielded from the pressure medium **104** by the housing container body **122**.

[0030] The lead frame **113** is provided for each predetermined external connection. A first end **113***a* of the lead frame **113** is exposed in a space inside the case body **111** while a second end **113***b* thereof protrudes in a connecting portion **111***d* of the case body **111**, from a side surface (surface substantially orthogonal to the bottom **111***a*) **111***c* of the case body **111**. Inside the case body **111**, the housing container body **122** is disposed so as to block the first open end **102***a* of the pressure intake part **102**. The housing container body **122** has a substantially concave shape opened in one portion in a cross-sectional view, the opened portion facing the pressure intake part **102** and being supported by a supporting portion **111***b* of the bottom **111***a* of the case body **111**.

[0031] A space (space bordered by an inner wall of the housing container body **122**) inside the housing container body **122** constitutes the pressure detection chamber **112**. The housing container body **122** spatially separates the pressure detection chamber **112** and another space (a space outside the housing container body **122**, hereinafter, wiring chamber) **114** inside the case body **111**. The first ends **113***a*, **123***a* of the lead frames **113**, **123** are connected to each other in the wiring chamber **114**. As described, the pressure medium **104** does not flow into the wiring chamber **114** and thus, configuration may be such that portions of the lead frames **113**, **123** exposed in the wiring chamber **114** are not encapsulated by an encapsulant **116**.

[0032] The housing container body **122** is a resin molded product in which the lead frame **123** is insert molded. A first end **123***a* of the lead frame **123** is exposed in the wiring chamber **114** while a second end **123***b* thereof is exposed in the pressure detection chamber **112**. Inside the housing container body **122**, a recess (hereinafter, sensor mounting portion) **124** that houses the pressure sensor chip **121** is formed. The pressure sensor chip **121** is die-bonded to a bottom **124***a* of the sensor mounting portion **124**, via a pedestal member **125** and an adhesive layer **126**, and is electrically connected to the second end **123***b* of the lead frame **123** by a bonding layer **127**.

[0033] The pressure sensor chip **121** is a semiconductor integrated circuit (IC) employing a piezoresistive method that utilizes the piezoresistive effect of a diffused resistor formed in a silicon (Si) semiconductor, the pressure sensor chip **121** being a pressure-sensitive surface type semiconductor integrated circuit (IC) having a gauge surface (circuit surface where a strain gauge is provided) that is a pressure detecting surface that detects pressure of the pressure medium **104**. The pressure sensor chip **121** has a diaphragm **121***a* that faces the first open end **102***a* of the pressure intake part **102**, and a back surface recess **121***b* that is blocked by the pedestal member **125**. The pressure sensor chip **121**, the bonding layer **127**, and the second end **123***b* of the lead frame **123** are encapsulated by an encapsulant **128** that is a gel type.

[0034] Operation of the semiconductor package **110** of the reference example is described. When the engine is running, combustion gas generated in the cylinders of the engine during a combustion and expansion process of the engine is discharged out of the automotive vehicle through the exhaust pipe as exhaust gas during the exhaust stroke. At this time, a portion of the exhaust gas flows into the pressure intake part **102** from the exhaust pipe and is sent to the pressure detection chamber **112** of the pressure sensor **101** as the pressure medium **104**. Via the encapsulant **128**, the diaphragm **121***a* of the pressure sensor chip **121** is distorted by pressure from by the pressure medium **104** and the pressure sensor **101** outputs an electrical signal corresponding to the distortion to an external circuit via the lead frames **123**, **113**.

[0035] Airtightness of the pressure intake part **102** and the exhaust pipe is increased by an O-ring **103** that blocks a gap between the pressure intake part **102** and the exhaust pipe; the pressure medium **104** that flows into the pressure intake part **102** from the exhaust pipe does not leak outside, reaches the pressure detection chamber **112**, and is sent to the pressure sensor chip **121**. A path of the pressure medium **104** from the exhaust pipe to the pressure detection chamber **112** is always spatially connected and no member is provided so as to shield the pressure sensor chip **121** from the pressure medium **104**. Thus, the thermal energy generated during the combustion process of the engine is transferred from the pressure medium **104** to the pressure sensor chip **121** substantially without being dissipated.

[0036] The pressure detection chamber **112** and the exhaust pipe are spatially connected in a straight line by the pressure intake part **102** and thus, the pressure medium **104** reaches the pressure sensor chip **121** substantially without being cooled. The pressure sensor chip **121** receives the most thermal energy from the pressure medium **104** and becomes the hottest among the components of the pressure sensor **101**. On the other hand, the lead frame **113** is formed of a metallic material and has high thermal conductivity. The lead frame **113** is connected to an external circuit and functions as a heat sink that receives the heat generated in the pressure sensor chip **121** from the lead frame **123** and releases the heat to the outside and thus, cools more easily than other components of the pressure sensor **101**.

[0037] The temperature of the lead frame **123** is lower than the temperature of the pressure sensor chip **121** and thus, the temperature of the encapsulant **128**, at a portion **142** thereof encapsulating the lead frame **123** is lower than a portion (portion where the pressure medium **104** is closest to the pressure sensor chip **121**) **141** thereof encapsulating the pressure sensor chip **121**. The pressure medium **104** contains vapor that is generated during the combustion of fuel (gasoline, etc.); the vapor penetrates the encapsulant **128**, is naturally cooled by the portion **142** of the encapsulant **128** having a relatively low temperature, whereby the vapor is condensed and discharged as condensate. Thus, the condensate may cause corrosion of the lead frame **123** and short circuit or disconnection of the bonding layer **127**.

[0038] Suppressing condensation in a pressure detection chamber is one issue solved by the present embodiments.

[0039] A semiconductor package according to an embodiment of the present disclosure is described in detail with reference to the accompanying drawings. In the description below and the accompanying drawings, main components that are identical are given the same reference numerals and are not repeatedly described.

[0040] A semiconductor package according to a first embodiment solving the problems above is described. FIG. **1** is a cross-sectional view schematically depicting a structure of a semiconductor package according to a first embodiment. In FIG. **1**, flow of a pressure medium **4** is indicated by a bold arrow (similarly in FIG. **3**). FIG. **2** is a plan view schematically depicting a layout when an inside of a case body depicted in FIG. **1** is viewed from a cover body. FIG. **2** depicts a bottom (a bottom **11***a* of the case body **11** and a top of a pressure detection chamber **12**) of the wiring chamber **14** of the case body **11**, and members disposed at the bottom of the wiring chamber **14**. In FIG. **2**, a connecting portion **11***d* for external connection of the case body **11** is not depicted.

[0041] A semiconductor package **10** according to the first embodiment depicted in FIGS. **1** and **2** includes a pressure sensor **1** and a pressure intake part **2** and is an on-board automotive sensor package that is for measuring pressure and installed to piping of an internal combustion engine (engine). The semiconductor package **10**, for example, is installed to exhaust system piping (exhaust pipe, not depicted) of the engine. The semiconductor package **10** is useful when the pressure medium **4** is a gas having a higher temperature and a higher humidity than the temperature and humidity of outside air and is particularly suitable for measuring the pressure of exhaust gas (exhaust pressure measurement). The semiconductor package **10** is not limited to exhaust pipes of automotive vehicles and may be installed to other piping (for example, turbocharger piping) through which a gas having a higher temperature and a higher humidity than the temperature and humidity of outside air.

[0042] The pressure sensor **1** is mounted to a pressure sensor chip (semiconductor substrate) **21** in the pressure detection chamber **12** inside the case body **11**, and outputs, as an electrical signal, pressure detected by the pressure sensor chip **21**, the electrical signal being output to an external circuit (not depicted) via a lead frame **13**. The pressure sensor **1** is supplied with voltage from a power source IC (not depicted) of an engine control unit (ECU) for controlling the engine or a power source IC (not depicted) of an electronic control unit (ECU) for controlling the driving performance, safety, and environmental friendliness of the automotive vehicle and is controlled by the ECU.

[0043] The case body **11** (including the connecting portion **11***d*), a later-described housing container body **22**, the pressure intake part **2**, and the exhaust pipe, for example, are formed of a high-performance resin having excellent heat resistance, mechanical strength, durability (chemical resistance, abrasion resistance), non-combustibility, electrical insulation, processability, etc., a so-called super engineering plastic such as polyphenylene sulfide (PPS). Thus, as compared to an instance in which these components of the exhaust system of the engine are formed of a metal, the suppression of corrosion due to the exhaust gas and a reduction in the weight of the exhaust system of the engine may be realized. The lead frames **13**, **23**, for example, contain phosphor bronze as a main material.

[0044] The case body **11** is a resin molded product in which the lead frame **13** is insert molded and, for example, has a substantially rectangular shape in a plan view. The case body **11** has a partially open and substantially concave shape in a cross-sectional view, and the partially open part is blocked by a cover body **15**. A space (space bordered by an inner wall (the bottom **11***a* and side surfaces) and a top (the cover body **15**) of the case body **11**) inside the case body **11** is divided into the pressure detection chamber **12** and the wiring chamber **14** by the later-described housing container body **22**. The pressure detection chamber **12** and the wiring chamber **14** are shielded from the outside air by the case body **11**. The pressure detection chamber **12** is spatially connected to the pressure intake part **2** via the wiring chamber **14** and is not directly connected to the pressure intake part **2**.

[0045] The wiring chamber **14** is a space (space outside the housing container body **22**) between the top of the pressure detection chamber **12** and the cover body **15** inside the case body **11**. The lead frames **13**, **23** are exposed in the wiring chamber **14**. Inside the wiring chamber **14**, first ends **13***a*, **23***a* of the lead frames **13**, **23** are connected to each other. Preferably, all portions of the lead frame **13** and the lead frame **23** connected to the lead frame **13** exposed in the wiring chamber **14** may be encapsulated by a general gel-type encapsulant **16**. The encapsulant **16** functions as a cap film ensuring airtightness of the lead frames **13**, **23** inside the wiring chamber **14**.

[0046] A supporting portion **11***b* that supports the housing container body **22** is provided at an inner wall at the bottom (portion facing the exhaust pipe) **11***a* of the case body **11**. The supporting portion **11***b* is a concave portion formed at the inner wall of the bottom **11***a* of the case body **11**. An end of a sidewall of an opening of the housing container body **22** is fitted into the supporting portion **11***b* as described hereinafter. Further, first and second channels **31**, **32** are provided at the bottom **11***a* of the case body **11**. The first and second channels **31**, **32** are channels of the pressure medium **4** sent to the pressure detection chamber **12** and spatially connect the pressure intake part **2** and the pressure detection chamber **12**. Shapes in a plan view, widths, and the number of the first and second channels **31**, **32** may be suitably set.

[0047] In particular, the first channel **31** is a through hole penetrating through the bottom **11***a* of the case body **11**, from a first end (closed end) **2***a* of the pressure intake part **2** to the wiring chamber **14**, in a vicinity of the lead frame **13** (i.e., vicinity of the encapsulant **16**) and couples the wiring chamber **14** and the pressure intake part **2**. Heat dissipated from the pressure medium **4** flowing through the first channel **3** may actively heat the lead frames **13**, **23**. The pressure medium **4** flowing through the pressure intake part **2** may be first flowed to the vicinity of the lead frame **13**, and a path from the closed end **2***a* of the pressure intake part **2** to the wiring chamber **14** may be suitably set. The first channel **31** may meander inside the bottom **11***a* of the case body **11** so as to face the lead frame **13**.

[0048] The second channel **32** is a through hole meandering inside the bottom **11***a* of the case body **11**, from the wiring chamber **14** to the pressure detection chamber **12** and couples the wiring chamber **14** and the pressure detection chamber **12**. Inside the wiring chamber **14**, heat from the pressure medium **4** is dissipated to the outside air via the cover body **15** of the case body **11** and the pressure medium **4** is cooled naturally. The second channel **32** may be provided in a vicinity of a sidewall (portion orthogonal to the bottom **11***a*) **11***c* of the case body **11**. In the second channel **32**, heat is easily dissipated from the pressure medium **4** to the outside air via the sidewall **11***c* of the case body **11**, and the pressure medium **4** is further cooled naturally while traveling from the wiring chamber **14** to the pressure detection chamber **12**.

[0049] By design, the first and second channels **31**, **32** are formed in a margin within the bottom **11***a* of the case body **11**, where members cannot be disposed. For example, as depicted in FIG. **2**, the case body **11** has, in a plan view, a substantially rectangular shape that almost exactly fits necessary members (such as the housing container body **22** and the lead frames **13** and **23**) on the inner wall of the bottom **11***a*. Thus, the margin of the bottom **11***a* of the case body **11**, for example, is assumed to be a gap between the sidewall **11***c* of the case body **11** and members inside the case body **11**, a space between an inner wall of the sidewall **11***c* of the case body **11** and the lead frames **13**, **23**, a gap between the lead frames **23** that are adjacent to each other and not encapsulated, and the like.
[0050] The connecting portion **11***d* for external connection of the case body **11** protrudes outside from the sidewall **11***c* of the case body **11**. The connecting portion **11***d* of the case body **11** is a plug that mates with a socket of a connector portion of cables (harness, not depicted) for wiring within the engine compartment, and has therein the lead frame **13** (three in FIG. **2**) for each predetermined external connection as plug pins (terminals). The external connections of the lead frame **13**, for example, are wiring connections for supplying voltage from the ECU's power supply IC to the pressure sensor chip **21**, wiring connections for inputting control signals from the ECU to the pressure sensor chip **21**, and wiring connections for external output from the pressure sensor chip **21**.
[0051] A thickness of the bottom **11***a* of the case body **11** is suitably set, thereby enabling a length of the first and second channels **31**, **32** to be suitably set. Further, the thickness of the bottom **11***a* of the case body **11** is suitably set, thereby enabling warming of the pressure detection chamber **12** by the pressure medium **4** in the pressure intake part **2** to be suppressed even when the pressure intake part **2** faces the pressure detection chamber **12** via the bottom **11***a* of the case body **11**. The case body **11** may be an existing integrally molded product in which the first and second channels **31** and **32** are formed and may have nearly a same shape as the existing integrally molded product. Thus, existing internal members may be used for the lead frame **13** and the case body **11**.
[0052] The first end **13***a* of the lead frame **13** is exposed in the wiring chamber **14** while the second end **13***b* thereof protrudes in the connecting portion **11***d* of the case body **11**, from the sidewall **11***c* of the case body **11**. Inside the case body **11**, the housing container body **22** is disposed on the inner wall of the bottom **11***a* of the case body **11**. The housing container body **22** has, in a cross-sectional view, a substantially concave shape that is partially opened. The housing container body **22** is supported by the bottom **11***a* of the case body **11** with the partially open portion (side opposite the bottom) thereof facing the bottom **11***a* of the case body **11** and the end of the sidewall (portion substantially orthogonal to the bottom) of the opening fitting into the supporting portion **11***b* of the bottom **11***a* of the case body **11**.
[0053] A space (space bordered by the inner wall of the housing container body **22**) inside the housing container body **22** constitutes the pressure detection chamber **12**. The housing container body **22** is a resin molded product in which the lead frame **23** is insert molded. The first end **23***a* of the lead frame **23** is exposed in the wiring chamber **14** while the second end **23***b* is exposed in the pressure detection chamber **12**. The lead frame **23** is one of multiple lead frames **23** and in the wiring chamber **14**, a different one of the lead frames **23** is connected to each of the lead frames **13** of the case body **11**. Inside the pressure detection chamber **12**, a concave portion (sensor mounting portion) **24** in which the pressure sensor chip **21** is housed so as to face the pressure detection chamber **12** is formed at the top of the pressure detection chamber **12** (inner wall of the housing container body **22**, at a bottom thereof). The sensor mounting portion **24**, for example, has a substantially circular shape in a plan view.
[0054] The pressure sensor chip **21** is a semiconductor IC employing a piezoresistive method that utilizes the piezoresistive effect of a diffused resistor (gauge resistor) formed in a silicon (Si) semiconductor, the semiconductor IC having a gauge surface (circuit surface where a strain gauge is provided) that is a pressure-sensitive surface that senses the pressure medium **4**, which is an

object to be detected. The pressure sensor chip **21** has a center portion and an outer peripheral portion and a diaphragm structure in which the center portion is etched from a back of the pressure sensor chip **21** thereby forming a recess **21***b* whereby a thickness of the pressure sensor chip **21** is thinner in the center portion than in the outer peripheral portion. The pressure sensor chip **21** includes a diaphragm (pressure-sensitive portion) **21***a* that bends in response to pressure, the strain gauge (not depicted), and computing circuity (not depicted) for amplifying and compensating output of the strain gauge. The diaphragm **21***a*, for example, has a substantially circular shape in a plan view.

[0055] The strain gauge contains a material (Si semiconductor) having a piezoresistive effect and is configured by multiple gauge resistors (not depicted) connected to a bridge, the strain gauge being provided on a front side of the pressure sensor chip **21**, facing the diaphragm **21***a*. The gauge resistors are each electrically connected to the lead frame **13** via a bonding layer **27** and a surface electrode (not depicted) on a front surface of the pressure sensor chip **21**. Distortion of the diaphragm **21***a* occurring when pressure from the pressure medium **4** is received, is converted by the strain gauge into an electrical signal of a magnitude proportional to the pressure (potential difference occurring among the bridge of the gauge resistors in proportion to the pressure), and the electrical signal is led out to the external circuit via the lead frame **13**.

[0056] The outer peripheral portion of the pressure sensor chip **21**, at a back surface thereof, for example, is electrostatically bonded (anode bonded) to a first surface of a pedestal member **25** so that the recess **21***b* at the back surface of the pressure sensor chip **21** is sealed by the pedestal member **25**. A second surface of the pedestal member **25** is die-bonded (fixed) to a floor **24***a* of the sensor mounting portion **24** via an adhesive layer **26**. The pressure sensor chip **21** and the pedestal member **25** are disposed apart from sidewalls of the sensor mounting portion **24**. The pedestal member **25**, for example, is a glass substrate formed of heat-resistant glass. The pressure sensor chip **21**, the bonding layer **27**, and the second end **23***b* of the lead frame **23** are encapsulated by a general, gel-type encapsulant **28**.

[0057] The pressure intake part **2** is a hollow, cylindrical resin pipe (piping made of resin) having a diameter smaller than a diameter of the exhaust pipe and couples the first channel **31** and the exhaust pipe. The pressure intake part **2** is integrally molded with the case body **11** and protrudes externally from the bottom **11***a* of the case body **11**. The pressure intake part **2** is spatially connected to the pressure detection chamber **12**, via the first channel **31**, the wiring chamber **14**, and the second channel **32**. A first end (hereinafter, closed end) **2***a* of the pressure intake part **2** is closed by the bottom **11***a* of the case body **11** so as to be not directly connected to the pressure detection chamber **12**. The closed end **2***a* of the pressure intake part **2** may face the pressure detection chamber **12** via the bottom **11***a* of the case body **11**.

[0058] A second end (open end) **2***b* of the pressure intake part **2** is inserted into a mounting hole formed at a curved surface (side surface) of the exhaust pipe, whereby the semiconductor package **10** is directly installed to the exhaust pipe. Exhaust gas flowing through the exhaust pipe when the engine is running (when the automotive vehicle is traveling, idling, etc.) flows into the pressure intake part **2** as the pressure medium **4**. The pressure intake part **2** is a channel for the pressure medium **4** that is sent to the pressure detection chamber **12**; the pressure intake part **2** is always spatially connected to the exhaust pipe. A length of the pressure intake part **2** is short and the pressure sensor **1** and the exhaust pipe are positioned relatively close to each other. The pressure intake part **2**, from a portion thereof facing the pressure detection chamber **12** bordered by the bottom **11***a* of the case body **11**, extends linearly toward the exhaust pipe.

[0059] The exhaust pipe is a hollow, cylindrical resin pipe through which the exhaust gas (combustion gas) flows during the exhaust stroke of the engine, the exhaust pipe coupling cylinders (not depicted) of the engine and the outside (outside of the automotive vehicle). When the engine is running, the exhaust gas constantly flows into the pressure intake part **2** from the exhaust pipe, passes through the pressure intake part **2**, the first channel **31**, the wiring chamber **14**, and the second channel **32**, to be sent to the pressure detection chamber **12** as the pressure medium **4**. An O-ring **3** is provided on the outer periphery of the pressure intake part **2** to seal a gap between the exhaust pipe and the pressure intake part **2**. The O-ring **3** ensures the airtightness between the exhaust pipe and the pressure intake part **2**.

[0060] Operation of the semiconductor package **10** according to the first embodiment is described. When the engine is running, air is taken into the engine and during a process of exhausting combustion gas that is generated in the cylinders of the engine during a combustion/expansion process of the engine, the combustion gas is exhausted as exhaust gas outside of the automotive vehicle, through the exhaust pipe. A portion of the exhaust gas flows from the exhaust pipe into the pressure intake part **2** as the pressure medium **4**, and from the pressure intake part **2**, reaches the pressure detection chamber **12** by passing through the first channel **31** of the bottom **11***a* of the case body **11**, the wiring chamber **14** of the case body **11**, and the second channel **32** of the bottom **11***a* of the case body **11**.

[0061] Heat is dissipated from the pressure medium **4** that flows into the first channel **31**, the heat being dissipated to the lead frame **13** through a thin resin part (the bottom **11***a* of the case body **11**) between the first channel **31** and the lead frame **13**. As a result, the lead frame **13**, and the lead frame **23** connected to the lead frame **13** can be actively heated. Thereafter, the pressure medium **4** flows into the wiring chamber **14** and the second channel **32**, and heat is dissipated from the pressure medium **4** to the outside air through the cover body **15** (and further from the sidewall **11***c* of the case body **11**), whereby the pressure medium **4** is naturally cooled.

[0062] The pressure medium **4** that has been naturally cooled flows into the pressure detection chamber **12** and reaches the pressure sensor chip **21**. Thus, in the pressure detection chamber **12**, a temperature of the encapsulant **28** in a portion **42** thereof encapsulating the lead frame **23** is higher than in a portion (portion where the pressure medium **4** comes closest to the pressure sensor chip **21**) **41** thereof encapsulating the pressure sensor chip **21**. As a result, vapor contained in the pressure medium **4** may be suppressed from forming a condensate at the portion **42** encapsulating the lead frame **23**, whereby corrosion of the lead frame **23**, short circuit and/or disconnection of the bonding layer **27** may be suppressed.

[0063] Further, the pressure medium **4** is naturally cooled and thus, vapor contained in the pressure medium **4** is condensed and discharged as condensate. As described, the pressure medium **4** is naturally cooled in the wiring chamber **14** (and further in the second channel **32**) and thus, vapor contained in the pressure medium **4** is condensed while the pressure medium **4** is enroute to the pressure detection chamber **12** thereby enabling the amount of vapor contained in the pressure medium **4** to be reduced (dehumidified). Vapor contained in the pressure medium **4** has been reduced when the pressure medium **4** flows into the pressure detection chamber **12** and thus, condensation in the pressure detection chamber **12** may be suppressed.

[0064] The pressure medium **4** that reaches the pressure detection chamber **12** is transmitted to the pressure sensor chip **21**. A resistance value of the strain gauge changes in response to pressure applied to the diaphragm **21***a* of the pressure sensor chip **21** by the pressure medium **4** and the pressure sensor **1** outputs the change in the resistance value of the strain gauge as an electrical signal to an external circuit. While the response speed of the pressure sensor **1** decreases the longer is the path of the pressure medium **4**, airtightness of the path of the pressure medium **4** from the exhaust pipe to the pressure detection chamber **12** is maintained and thus, measurement accuracy of the pressure sensor **1** is about a same as the measurement accuracy of the pressure sensor **101** (refer to FIGS. **4** and **5**) of the reference example.

[0065] As described, according to the first embodiment, the pressure intake part and the pressure detection chamber are not directly connected but are spatially connected via the first and second channels formed inside the bottom of the case body of the pressure sensor. The first channel is formed in a vicinity of the lead frame, which is insert molded in the case body. The exhaust gas, which has a high temperature and a high humidity and flows from the exhaust pipe into the pressure intake part, flows as the pressure medium into the first channel from the closed end of the pressure intake part, passes a vicinity of the lead frame, and is sent to the pressure detection chamber. The lead frame may be actively heated by the pressure medium flowing through the first channel. As a result, the temperature of the lead frame may be raised to be higher than the temperature of the pressure sensor chip.

[0066] Further, the pressure medium is naturally cooled through a space (the wiring chamber and the second channel) different from the pressure detection chamber inside the case body before reaching the pressure detection chamber and the amount of vapor contained in the pressure medium is reduced. The naturally cooled pressure medium flows into the pressure detection chamber and is sent to the pressure sensor chip, whereby in the pressure detection chamber, the temperature of the encapsulant at a portion of the encapsulant encapsulating the lead frame is higher than the temperature of a portion of the encapsulant encapsulating the pressure sensor chip. As a result, condensation at the portion encapsulating the lead frame may be suppressed and thus, corrosion of the lead frame, short circuit and disconnection of the bonding layer may be suppressed. Further, the pressure medium is naturally cooled, whereby the amount of vapor contained in the pressure medium is reduced thereby enabling condensation in the pressure detection chamber to be suppressed.

[0067] A semiconductor package according to a second embodiment solving the problems above is described. FIG. **3** is a cross-sectional view schematically depicting a structure of the semiconductor package according to the second embodiment. A semiconductor package **50** according to the second embodiment differs from the semiconductor package **10** according to the first embodiment in that instead of the first and second channels **31**, **32** (refer to FIGS. **1** and **2**), first and second channels **51**, **52** constituting channels of a pressure medium **54** and a space **53** for heat transfer are provided. The first and second channels **51**, **52** and the space **53** for heat transfer are formed inside the bottom **11***a* of the case body **11**, at positions close to the pressure intake part **2**, the lead frame **13**, and the pressure detection chamber **12**.

[0068] In particular, the first channel **51** is a through hole reaching the space **53** for heat transfer from the closed end **2***a* of the pressure intake part **2** and couples the space **53** and the pressure intake part **2**. Preferably, the first channel **51** may be formed over a shortest distance from the closed end **2***a* of the pressure intake part **2** to the space **53**. The second channel **52** is a through hole that reaches the pressure detection chamber **12** from the space **53** for heat transfer and couples the space **53** and the pressure detection chamber **12**. The second channel **52** may be formed over a shortest distance from the space **53** to the pressure detection chamber **12**. The first and second channels **51** and **52** may meander through the bottom **11***a* of the case body **11** so as to face the lead frame **13**.

[0069] The space **53** for heat transfer is formed in the bottom **11***a* of the case body **11**, at a position close to the lead frame **13**. Preferably, the space **53** for heat transfer may face the lead frame **13** (preferably, all the lead frames **13**) in a direction orthogonal to the bottom **11***a* of the case body **11**. Heat from the pressure medium **54** that flows into the space **53** from the pressure intake part **2** via the first channel **51** is dissipated to the lead frame **13** via the thin resin portion (the bottom **11***a* of the case body **11**) between the space **53** and the lead frame **13**. Thus, the lead frames **13**, **23** can be actively heated by the pressure medium **54** in the space **53**.

[0070] In the second embodiment, the space (space bordered by the inner wall of the case body **11** and the cover body **15**) inside the case body **11** is spatially separated into the pressure detection chamber **12** and the wiring chamber **14** by the housing container body **22**. The pressure detection chamber **12** and the wiring chamber **14** are shielded from the outside air by the case body **11**. The pressure detection chamber **12** is spatially connected to the pressure intake part **2** via the space **53** in the bottom **11***a* of the case body **11** and is not directly connected to the pressure intake part **2**. The wiring chamber **14** is an enclosed space shielded from the outside air by the case body **11** and the cover body **15** and shielded from the pressure medium **54** by the housing container body **22**.

[0071] In other words, when the engine is running, a portion of the exhaust gas passes through the pressure intake part **2**, the first channel **51**, the space **53**, and the second channel **52** of the bottom **11***a* of the case body **11** from the exhaust pipe as the pressure medium **54** and reaches the pressure detection chamber **12**. Similar to the first embodiment, the lead frames **13**, **23** can be actively heated by the pressure medium **54** that flows into the space **53**.
[0072] Thereafter, the pressure medium **54** is naturally cooled enroute to the pressure sensor chip **21**. Thus, similar to the first embodiment, in the pressure detection chamber **12**, the temperature of the encapsulant **28** is higher at the portion **42** encapsulating the lead frame **23** than at the portion **41** encapsulating the pressure sensor chip **21**.
[0073] As described, according to the second embodiment, even when configuration is such that the pressure medium reaches the pressure detection chamber from the pressure intake part through the heat transfer space in the bottom of the case body, the same effect as the first embodiment may be obtained.
[0074] In the foregoing, the present disclosure is not limited to the embodiments described above and may be variously modified within a range not departing from the spirit of the present disclosure.
[0075] The semiconductor package according to the present disclosure achieves an effect in that condensation may be suppressed.
[0076] As described, the semiconductor package according to the present disclosure is useful for semiconductor packages that are for measuring pressure and installed to piping and is particularly suitable in instances when installed to exhaust system piping (exhaust pipe) of an internal combustion engine (engine) to measure the pressure of exhaust gas.
[0077] Although the invention has been described with respect to a specific embodiment for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art which fairly fall within the basic teaching herein set forth.

Claims

1. A semiconductor package for detecting a pressure of a pressure medium flowing through a pipe, the semiconductor package comprising: a case body having a pressure detection chamber; a pressure sensor chip housed in the pressure detection chamber and configured to convert a pressure applied thereto into an electrical signal; a lead frame insert-molded in the case body and electrically connected to the pressure sensor chip; and a pressure intake part having: a first end protruding externally from a bottom of the case body, and a second end spatially connected to the pipe, wherein the case body has a channel formed in the bottom thereof coupling the pressure detection chamber and the pressure intake part, the channel extending from the first end of the pressure intake part and passing through a vicinity of the lead frame, to thereby guide the pressure medium from the first end of the pressure intake part to the pressure detection chamber, such that the pressure medium applies the pressure to the pressure sensor chip in the pressure detection chamber.
2. The semiconductor package according to claim 1, wherein the case body has a space different from the pressure detection chamber, and the pressure medium is guided from the first end of the pressure intake part through both the channel and the space to the pressure detection chamber.
3. The semiconductor package according to claim 2, wherein the channel includes: a first channel passing through the vicinity of the lead frame and coupling the pressure intake part and the space, and a second channel coupling the space and the pressure detection chamber.
4. The semiconductor package according to claim 3, wherein the space is partially formed by a top of the case body.
5. The semiconductor package according to claim 3, wherein the second channel is partially formed by a side surface of the case body.

6. The semiconductor package according to claim 2, wherein the space is formed inside the bottom of the case body, and in the vicinity of the lead frame, and the channel includes: a first channel coupling the pressure intake part and the space, and a second channel coupling the space and the pressure detection chamber.
7. The semiconductor package according to claim 6, wherein the space faces the lead frame in a direction orthogonal to the bottom of the case body.
8. The semiconductor package according to claim 1, wherein the pressure medium is gas, and the pipe is an engine pipe.